Lesson 1.2 - Algebraic Expression

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Objectives

At the end of the lesson, the students will be able to:

- Define algebraic expressions.
- Simplify algebraic expressions.
- Perform operations on algebraic expressions.
- Define polynomials.
- Factor polynomials.
- Simplify rational expressions and perform operations on rational expressions.
- Simplify radical expressions and perform operations on radical expressions.

Algebraic Expressions

A variable is a letter that can represent any number from a given set of numbers. If we start with variables, such as x, y, and z, and some real numbers and combine them using addition, subtraction, multiplication, division, powers, and roots, we obtain an algebraic expression. Here are some examples:

$$2x^2 - 3x + 4, \quad \sqrt{x} + 10, \quad \frac{y - 2z}{y^2 + 4}.$$

A monomial is an expression of the form ax^k, where a is a real number and k is a nonnegative integer. A binomial is a sum of two monomials and a trinomial is a sum of three monomials. In general, a sum of monomials is called a polynomial. For example, the first expression listed above is a polynomial, but the other two are not.

Polynomials

Definition

A polynomial in the variable x is an expression of the form $a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$ where $a_0, a_1, \ldots, a_n$ are real numbers and n is a nonnegative integer. If $a_n \neq 0$, then the polynomial has degree n. The monomials $a_k x^k$ that make up the polynomial are called the terms of the polynomial.

Polynomial	**Type**	**Terms**	**degree**
$2x^2 - 3x + 4$	trinomial	$2x^2, -3x, 4$	2
$x^8 + 5x$	binomial	$x^8, 5x$	8
$8 - x + x^2 - \frac{1}{2}x^3$	four terms	$8, -x, x^2, -\frac{1}{2}x^3$	3
$5x + 1$	binomial	$5x, 1$	1
$9x^5$	monomial	$9x^5$	5
6	monomial	6	0

Table: Examples of polynomials

Addition/Subtraction of Polynomials

We add and subtract polynomials using the properties of real numbers that were discussed in Lesson 1.1. The idea is to combine like terms (that is, terms with the same variables raised to the same powers) using the Distributive Property. For instance,

$$5x^7 + 3x^7 = (5+3)\,x^7 = 8x^7.$$

Example

1. Find the sum $\left(x^3 - 6x^2 + 2x + 4\right) + \left(x^3 + 5x^2 - 7x\right)$.
2. Find the difference $\left(x^3 - 6x^2 + 2x + 4\right) - \left(x^3 + 5x^2 - 7x\right)$.

Note: The presence of certain symbols of grouping in algebraic expressions indicates that certain operations are to be performed in a specified order.

Grouping symbols

Steps in removing symbols of grouping:

1. If a compound symbol of grouping exists, start removing the innermost symbol.
2. If a term (numerical coefficient, literal coefficient, or both) appears before a symbol of grouping, multiply this term to every term inside the symbol.
3. If a positive (+) sign preceded a symbol, the symbol may be removed without any changes done to the enclosed terms.
4. If a negative (−) sign proceeds a symbol, the symbols may be removed, and change the sign of every enclosed term.
5. Simplify the expression by combining like terms.

Example

1. $a - b + 2(3a - 4b) - (-3a + 8b)$
2. $3x - \{4x - 2y - 3[4x - 3y + z - (2x - 4y + 3z) - 2x] + 4z\} - 3y$

Theorem

The following are the fundamental laws on exponents:

1. $a^0 = 1$
2. $a^m a^n = a^{m+n}$
3. $(a^m)^n = a^{mn}$
4. $(ab)^n = a^n b^n$
5. $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
6. $\frac{a^m}{a^n} = a^{m-n}$
7. $a^{-n} = \frac{1}{a^n}$
8. $a^{\frac{1}{n}} = \sqrt[n]{a}$
9. $a^{\frac{m}{n}} = \left(a^{\frac{1}{n}}\right)^m = (\sqrt[n]{a})^m$ *or*
   $a^{\frac{m}{n}} = (a^m)^{\frac{1}{n}} = \sqrt[n]{a^m}$
10. $a^{-\frac{m}{n}} = \frac{1}{a^{\frac{m}{n}}}$

Laws of exponents

Example

Simplify and reduce the following exponential expressions:

1. $\dfrac{64r^3s^6p^5}{56r^5s^4p^3}$
2. $\dfrac{7b^4c^3}{8d^2} \cdot \dfrac{16b^3d^3}{21c^4d^5}$
3. $\left[\dfrac{4r^2s^3}{3t^4}\right]^3 \left[\dfrac{9s^2t^4}{8r^3}\right]^4$
4. $\left[\dfrac{9^{-1}m^{-2}g^2}{6^{-3}m^3g^3}\right]^2$
5. $\dfrac{a^{-2}+b^{-2}}{a^{-2}-b^{-2}}$

Definition

The product of polynomials is obtained by the use of the commutative, associative, and distributive axioms, the law of signs, and the law of exponents in multiplication. In particular, using it three times on the product of two binomials, we get

$$(a+b)(c+d) = a(c+d) + b(c+d) = ac + ad + bc + bd.$$

Example

1. $3a^4 \cdot 8a^3$
2. $2xy^3\,(-3x^2y)$
3. $3x^2yz^3\,(-2x^3y^2z)\,(6xy^4z^5)$
4. $3ab\,(2a - 4b + 7a2b)$
5. $(3a + 2b)\,(a - 3b)$
6. $(p - 4q + r)\,(3p + q - 2r)$

Special Products and factoring

You will recall that when two or more numbers or algebraic expressions are multiplied to form a product, they are called the factors of the product. When we start with a product and break it up into its factors, the process is called factoring.

SPECIAL PRODUCT FORMULAS

If A and B are any real numbers or algebraic expressions, then

1. $(A+B)(A-B) = A^2 - B^2$ — Sum and difference of same terms
2. $(A+B)^2 = A^2 + 2AB + B^2$ — Square of a sum
3. $(A-B)^2 = A^2 - 2AB + B^2$ — Square of a difference
4. $(A+B)^3 = A^3 + 3A^2B + 3AB^2 + B^3$ — Cube of a sum
5. $(A-B)^3 = A^3 - 3A^2B + 3AB^2 - B^3$ — Cube of a difference

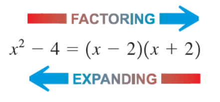
Figure: Special products formulas

Example

Expand the following expressions

1. $(3x+5)^2$
2. $(x^2-2)^3$
3. $(2x-\sqrt{y})(2x+\sqrt{y})$
4. $(x+y-1)(x+y+1)$

We use the Distributive Property to expand algebraic expressions. We sometimes need to reverse this process (again using the Distributive Property) by factoring an expression as a product of simpler ones. For example, we can write



Factoring common factors

We say that $x - 2$ and $x + 2$ are factors of $x^2 - 4$. The easiest type of factoring occurs when the terms have a common factor.

Example

Factor each expression

1. $3x^2 - 6x$
2. $ab + ac - ad$
3. $8x^4y^2 + 6x^3y^3 - 2xy^4$
4. $(2x + 4)(x - 3) - 5(x - 3)$
5. $a(x^2 + y^2) - a(x^2 - xy - y^2)$

To factor a trinomial of the form $x^2 + bx + c$, we note that

$$(x + r)(x + s) = x^2 + (r + s)x + rs$$

so we choose numbers r and s so that $r + s = b$ and $rs = c$.

Example

Factor $x^2 + 7x + 12$.

To factor a trinomial of the form $ax^2 + bx + c$ with $a \neq 1$, we look for factors of the form $px + r$ and $qx + s$:

$$ax^2 + bx + c = (px + r)(qx + s) = pqx^2 + (ps + qr)x + rs.$$

Therefore we try to find numbers p, q, r, and s such that $pq = a$, $rs = c$, $ps + qr = b$. If these numbers are all integers, then we will have a limited number of possibilities to try for p, q, r, and s.

Example

Factor $6x^2 + 7x - 5$.

Factoring Trinomials

Example (Recognizing the form of an expression)

Factor each expression

1. $x^2 - 2x - 3$
2. $(5a+1)^2 - 2(5a+1) - 3$
3. $15x^2 + 2x - 8$
4. $20m^2 + 3mn - 35n^2$
5. $3y^3 + 2y^2 - 5y$

Special factoring formulas

SPECIAL FACTORING FORMULAS

Formula	Name
1. $A^2 - B^2 = (A - B)(A + B)$	Difference of squares
2. $A^2 + 2AB + B^2 = (A + B)^2$	Perfect square
3. $A^2 - 2AB + B^2 = (A - B)^2$	Perfect square
4. $A^3 - B^3 = (A - B)(A^2 + AB + B^2)$	Difference of cubes
5. $A^3 + B^3 = (A + B)(A^2 - AB + B^2)$	Sum of cubes

Some examples

Example

Factor each expression

1. $9x^2 + 24x + 16$
2. $4x^2 - 25$
3. $(x + y)^2 - z^2$
4. $x^2 + 6x + 9$
5. $4x^2 - 4xy + y^2$
6. $4a^2 - 4a + 1$
7. $27x^3 - 1$
8. $x^6 + 8$

General suggestions for factoring

Usually, a student finds it comparatively easy to factor a list of exercises, which are all of one type form. A list of exercises involving all the types studied presents more difficulty. The following suggestions for factoring a varied list of exercises will prove helpful:

1. First look for a common monomial factor (other than 1) and separate the expression into its monomial factor and the corresponding polynomial factor.
2. Study the polynomial factor after the monomial factors are removed. Determine to which type of factoring it should be classed and then use the methods of factoring applicable to that type.
3. Continue with step 2 until all the factors obtained are prime factors.

Factoring completely

An algebraic expression may have more than two factors. When two factors have been found, each should be inspected to see whether further factoring is possible. With proper groupings, an expression can also be factored applying the different types of factoring we have already studied.

Example

Factor each expression completely.

1. $6a^3b + 3a^2b^2 - 18ab^3$
2. $4x^3 - 12x^2 - x + 3$
3. $a^2 + ab - 2b^2 + 2a - 2b$
4. $2x^4 - 8x^2$
5. $x^5y^2 - xy^6$
6. $x^3 - 2x^2 - 9x + 18$
7. $3x^{3/2} - 9x^{1/2} + 6x^{-1/2}$
8. $(2 + x)^{-2/3}x + (2 + x)^{1/3}$

Rational expressions

Definition

A quotient of two algebraic expressions is called a fractional expression, e.g.,

$$\frac{2x}{x-1}, \quad \frac{y-2}{y^2+4}, \quad \frac{x^3-x}{x^2-5x+6}, \quad \frac{x}{\sqrt{x^2+1}}.$$

Definition

A rational expression, which is another term for fraction, is a fractional expression in which both the numerator and the denominator are polynomials.

The following are other examples of rational expressions:

$$\frac{4a}{7}, \quad \frac{3x^2}{2xy}, \quad \frac{c^2-d^2}{c^2+d^2}, \quad \frac{(x+y)(x^4+xy+y^4)}{(x+y)(x^3+x^2y+y^2)}$$

Note: $\dfrac{x}{\sqrt{x^2+1}}$ is not a rational expression.

Rational expressions

In dealing with fractions, it is worthwhile to recall the laws on exponents, operations involved in special products and factoring as well as to familiarize the following theorem:

Theorem

If $\dfrac{a}{b} = \dfrac{c}{d}$, *then* $ac = bd$ *provided that* b *and* d *are nonzero.*

Theorem

1. $\dfrac{-a}{b} = \dfrac{a}{-b} = -\dfrac{a}{b}$, *provided that* $b \neq 0$.
2. $\dfrac{a}{b} = \dfrac{-a}{-b} = -\dfrac{-a}{b} = -\dfrac{a}{-b}$, *provided that* $b \neq 0$.
3. $\dfrac{a}{b} = \dfrac{a \cdot n}{b \cdot n}$, *for* $n \neq 0$.

Simplifying rational expressions

A rational expression is said to be in its lowest terms if both the numerator and denominator have no common factors except 1. Such a fraction is called a reduced fraction. For example, $24/36 = 2/3$, and $2/3$ is the reduced form of $24/36$ since 2 and 3 have no common factors other than 1.

In reducing a given fraction or a rational expression into its lowest terms, divide both the numerator and denominator by the product of the factors that are common to both and obtain the numerator and denominator, respectively, of the reduced fraction. If the common factors are not clearly visible, it is advisable to factor both the numerator and denominator first before reducing the given fraction.

> To simplify rational expressions, we factor both numerator and denominator and use the following property of fractions:
>
> $$\frac{AC}{BC} = \frac{A}{B}.$$
>
> This allows us to cancel common factors from the numerator and denominator.

Some examples

Example

Simplify:

1. $\dfrac{35a^4b^2}{42a^3b^3}$
2. $\dfrac{x^2-1}{x^2+x-2}$
3. $\dfrac{x^3+x^2-6x}{x^3-3x^2+2x}$
4. $\dfrac{2x^3-x^2-6x}{2x^2-7x+6}$
5. $\dfrac{1-x^2}{x^3-1}$

Multiplying and dividing rational expressions

Definition

The product of two or more rational expressions is a rational expression whose numerator is equal to the product of the given numerators and whose denominator is the product of all the given denominators, e.g.,

$$\frac{A}{B} \cdot \frac{C}{D} = \frac{AC}{BD}.$$

If possible, factor all the members of the given fractions before the product is formed. The factors that are common both in the numerator and denominator can just be canceled out easily.

Example

Perform the indicated multiplication and simplify:

1. $\dfrac{x^2+2x-3}{x^2+8x+16} \cdot \dfrac{3x+12}{x-1}$
2. $\dfrac{x^2+2x-15}{x^2-25} \cdot \dfrac{x-5}{x+2}$
3. $\dfrac{x^2+2xy+y^2}{x^2-y^2} \cdot \dfrac{2x^2-xy-y^2}{x^2-xy-2y^2}$
4. $\dfrac{a^2-4b^2}{2a^2-7ab+3b^2} \cdot \dfrac{6a-3b}{2a+4b} \cdot \dfrac{a^2-4ab+3b^2}{a^2-ab-2b^2}$
5. $\dfrac{x^2-3x+2}{2x^2+3x-2} \cdot \dfrac{2x^2+5x-3}{x^2-1} \cdot \dfrac{3x^2+6x}{2x-4}$
6. $\dfrac{x^2-x-6}{x^2+2x} \cdot \dfrac{x^3+x^2}{x^2-2x-3}$

Multiplying and dividing rational expressions

Theorem

To divide rational expressions, we use the following property of fractions:

$$\frac{A}{B} \div \frac{C}{D} = \frac{A}{B} \cdot \frac{D}{C}.$$

This says that to divide a fraction by another fraction, we invert the divisor and multiply.

Example

Perform the indicated division and simplify:

1. $\dfrac{x-4}{x^2-4} \div \dfrac{x^2-3x-4}{x^2+5x+6}$
2. $\dfrac{x^2-3x+2}{2x^2-7x+3} \div \dfrac{x^2-x-2}{2x^2+3x-2}$
3. $\dfrac{\dfrac{x^3}{x+1}}{\dfrac{x}{x^2+2x+1}}$

Compound/Complex fractions

A complex fraction is a fraction in which one or more of the terms of one or both members is a fraction. The following are examples of complex fractions:

$$\frac{3}{\frac{2}{3}}, \quad \frac{1+\frac{x}{y}}{x+y}, \quad \frac{\frac{4x}{x+y}+\frac{2y}{x-y}}{3-\frac{x^2+y^2}{x^2-y^2}}$$

A complex fraction can be simplified by reducing both the numerator and denominator first to single fractions before dividing the numerator by the denominator.

Example

Simplify the compound fractional expression.

1. $$\frac{x + \dfrac{y}{x}}{y + \dfrac{x}{y}}$$

2. $$\frac{1 + \dfrac{1}{1 + \dfrac{1}{x-1}}}{\dfrac{1}{1 - \dfrac{1}{x+1}}}$$

3. $$\frac{x^{-2} - y^{-2}}{x^{-1} + y^{-1}}$$

Addition/Subtraction of rational expressions

The sum of two or more fractions having the same denominator is a fraction with the same common denominator as its denominator and the sum of the numerators as its numerator. Thus,

$$\frac{A}{C} + \frac{B}{C} = \frac{A+B}{C}.$$

Example

Find the sum of the following:

1. $\dfrac{2a}{a-b} - \dfrac{6b}{a-b} + \dfrac{a+2b}{a-b}$
2. $\dfrac{1}{u+1} + \dfrac{u}{u+1}$

To find the algebraic sum of two or more fractions with different denominators, the least common denominator must have to be determined first.

The process of finding the algebraic sum of fractions having different denominators consists of the following steps.

1. Factor the given denominators into prime factors.
2. Determine the least common denominator (LCD). The LCD becomes the denominator of the resulting fraction.
3. Divide the LCD by the denominator of the first fraction and multiply the quotient to the numerator of the first fraction. The product becomes the first term of the numerator of the resulting fraction.
4. Repeat step 3 for the rest of the given fractions.
5. Simplify and determine the algebraic sum of the terms in the numerator of the resulting fraction.

Addition/Subtraction of rational expressions

Example

Perform the indicated operations and simplify.

1. $\dfrac{3}{x-1} + \dfrac{x}{x+2}$
2. $\dfrac{1}{x^2-1} - \dfrac{2}{(x+1)^2}$
3. $\dfrac{x^2-2xy}{3(x^2-y^2)} + \dfrac{y}{6(x-y)} - \dfrac{x}{4(x+y}$
4. $\dfrac{1}{x^2+3x+2} - \dfrac{1}{x^2-2x-3}$
5. $\dfrac{1}{x+1} - \dfrac{2}{(x+1)^2} + \dfrac{3}{x^2-1}$

Addition/Subtraction of rational expressions

Don't make the mistake of applying the properties of multiplication to the operation of addition. Many of the common errors in algebra involve doing just that. The following table states several properties of multiplication and illustrates the error in applying them to addition.

Correct multiplication property	**Common error with addition**
$(a \cdot b)^2 = a^2 \cdot b^2$	$(a + b)^2 \neq a^2 + b^2$
$\sqrt{a \cdot b} = \sqrt{a}\sqrt{b} \quad (a, b \geq 0)$	$\sqrt{a + b} \neq \sqrt{a} + \sqrt{b}$
$\sqrt{a^2 \cdot b^2} = a \cdot b \quad (a, b \geq 0)$	$\sqrt{a^2 + b^2} \neq a + b$
$\frac{1}{a} \cdot \frac{1}{b} = \frac{1}{a \cdot b}$	$\frac{1}{a} + \frac{1}{b} \neq \frac{1}{a + b}$
$\frac{ab}{a} = b$	$\frac{a + b}{a} \neq b$
$a^{-1} \cdot b^{-1} = (a \cdot b)^{-1}$	$a^{-1} + b^{-1} \neq (a + b)^{-1}$

Radical expressions

Definition

The principal n^{th} root of a real number a, $n \geq 2$ an integer, symbolized by $\sqrt[n]{a}$, is defined as follows:

$$\sqrt[n]{a} = b \text{ means } a = b^n,$$

where $a \geq 0$ and $b \geq 0$ if n is even and a, b are any real numbers if n is odd.

Note: The symbol $\sqrt[n]{a}$ for the principal n^{th} root of a is called a radical; the integer n is called the index, and a is called the radicand. If the index of a radical is 2, we call $\sqrt[2]{a}$ a the square root of a and omit the index 2 by simply writing $\sqrt{a}$.

For examples:

$$\sqrt{4} = 2, \quad \sqrt[3]{8} = 2, \quad \sqrt[5]{-243} = -3$$

Therefore, $a^{\frac{1}{n}}$ is defined as the principal n^{th} root of a. Hence, $a^{\frac{1}{n}} = \sqrt[n]{a^1}$. Generally,

$$\boxed{a^{\frac{r}{n}} = (\sqrt[n]{a})^r = \sqrt[n]{a^r}.}$$

In many cases, it is more advantageous to express a quantity in terms of radicals than in terms of rational exponents. The laws of radicals follow directly from the previous definition and theorems on exponents. If m and n are positive integers and a and b are also positive:

- $(\sqrt[n]{a})^n = a$
- $\sqrt[n]{ab} = (ab)^{\frac{1}{n}} = a^{\frac{1}{n}} \cdot b^{\frac{1}{n}} = \sqrt[n]{a}\,\sqrt[n]{b}$
- $\sqrt[n]{\dfrac{a}{b}} = \left(\dfrac{a}{b}\right)^{\frac{1}{n}} = \dfrac{a^{\frac{1}{n}}}{b^{\frac{1}{n}}} = \dfrac{\sqrt[n]{a}}{\sqrt[n]{b}}$
- $\sqrt[n]{\sqrt[m]{a}} = (a^{\frac{1}{m}})^{\frac{1}{n}} = a^{\frac{1}{mn}} = \sqrt[mn]{a}$

Simplification of radical expressions

A complete simplification of radicals by the use of the laws on radicals will yield:

1. no factors which are perfect n^{th} powers under a radical whose index is n
2. no fraction under a radical sign or no radical appears in the denominator, i.e., the denominator is rationalized.
3. the smallest possible index of the radical

Any radical that satisfies the above conditions is said to be in its simplest form.

Example

Simplifying Radicals

1. $\sqrt{32}$
2. $\sqrt[3]{16}$
3. $\sqrt{192a^3b^7}$
4. $\sqrt[3]{-16x^4}$
5. $\sqrt[4]{\frac{16x^5}{81}}$
6. $\sqrt[3]{81x^5y^7}$

Rationalizing radical expressions

Using the laws on radicals, it is always possible to convert a fraction whose denominator is a monomial or a binomial that contains a radical expression to an equal fraction in which the denominator contains no radical expression. This process is called rationalizing the denominator. In many computations involving radical expressions, it is advisable to rationalize all denominators as a first step.

> An expression that when multiplied to a radical expression yields an expression free of a radical sign is called a rationalizing factor or conjugate.

Expression	Rationalizing factor or conjugate
$\sqrt{3}$	$\sqrt{3}$
$\sqrt{8}$	$\sqrt{2}$
$\sqrt[3]{4}$	$\sqrt[3]{2}$
$\sqrt{a}+\sqrt{b}$	$\sqrt{a}-\sqrt{b}$
$5+\sqrt{2}$	$5-\sqrt{2}$

Note: The use of the least rationalizing factor should be observed to avoid yielding an expression not in the lowest term.

Example (Monomial denominators)

Rationalize the denominator of each expression:

1. $\dfrac{2}{\sqrt{3}}$
2. $\dfrac{5}{4\sqrt{2}}$
3. $\sqrt[3]{\dfrac{3a}{2b^2c}}$
4. $\dfrac{x^2-y^2}{2x\sqrt{x+y}}$

Multiplication/Division of radical expressions

Obtaining the product and quotient of radical expressions is governed by the second and third laws on radicals.

If the radicals have the same index, their radicands can be multiplied and placed under a single radical sign. However, if the radicals are of different indexes, they must be converted first to radicals having similar indices before the multiplication of the radicands can take place. This is always possible with the aid of equivalent expressions having rational exponents.

Division of radicals is handled in a similar manner. Again, the radicals must have similar indices first before the division of the radicals can be carried out.

Multiplication/Division of radical expressions

Example (Multiplication)

1. $\sqrt{14}\sqrt{21}$
2. $\sqrt{15ax^3}\sqrt{45a^2xy^3}$
3. $\sqrt{(x^2-y^2)}\sqrt{(x-y)}$
4. $(\sqrt{5}+2\sqrt{3})(\sqrt{5}-3\sqrt{3})$
5. $\sqrt{6x^3}\sqrt[3]{4x^4y^2}$

Example (Division)

1. $\dfrac{\sqrt{18}\sqrt{12}}{\sqrt{6}}$
2. $\dfrac{\sqrt[4]{24a^3b}}{\sqrt[4]{8ab^3}}$
3. $\dfrac{(x+y)\sqrt{(x+y)}}{\sqrt{(x^2-y^2)}}$
4. $\dfrac{\sqrt[6]{12}}{\sqrt{3}\sqrt[3]{2}}$

Addition and subtraction of radical expressions

In adding or subtracting radical expressions, all similar radicals (that is, those containing the same index and radicand) are combined into single terms.

If the radical expressions to be added or subtracted have different orders or radicands, the following procedure is suggested.

1. Simplify each radical expression and rationalize all denominators.
2. If it is possible to convert one or more radical expressions to a lower order, do it.
3. If possible, assemble the resulting radical expressions into one or more groups composed of radical expressions of the same or radicands then apply the distributive axiom to each group.

Addition and subtraction of radical expressions

Example

1. $5\sqrt{a} + 3\sqrt{a} - 2\sqrt{a}$
2. $\sqrt{108} + \sqrt{48} - \sqrt{3}$
3. $\sqrt[3]{a^4 b} + \dfrac{1}{\sqrt[3]{a^2 b^2}} + 3\sqrt[3]{ab^4}$
4. $8\sqrt{\dfrac{1}{3}} + \dfrac{3}{2}\sqrt{108} - \sqrt[4]{9}$
5. $\sqrt{8a^3 b^3} + \sqrt[3]{ab} - \sqrt{\dfrac{2}{ab}} - \sqrt[3]{8a^4 b^4} - \sqrt[4]{4a^2 b^2}$